

Extra Credit Assignment

Arham Lodha

May 01, 2024

Claim 1. Let Ω be a open set that contains the closure of the disk D_R and suppose f is holomorphic on Ω , $f(0) \neq 0$, and f vanishes nowhere on C_R . If $z_1, \dots, z_N$ are the zeros of f inside the disks (counted with multiplicities, ie roots can be repeated), then,

$$\log |f(0)| = \sum_{k=1}^N \log \left(\frac{|z_k|}{R} \right) + \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta \quad (1)$$

Proof. There are 4 steps to solve this theorem.

Step 1: Note that $\forall f_1, f_2$ which satisfy the hypothesis, $f_1 f_2$ satisfy the hypothesis. Moreover if f_1 and f_2 satisfy the conclusion of the theorem, $f_1 f_2$ also satisfies the conclusion because $|f_1(x)|, |f_2(x)| \in \mathbb{R}$ and $\log(xy) = \log(x) + \log(y)$ for all $x, y \in \mathbb{R}^+$ and the fact that the zeros of $f_1 f_2$ are the union of the zeros of f_1 and f_2 . So we can reduce to a set of base functions.

Step 2: The function:

$$g(z) = \frac{f(z)}{\prod_{i=1}^N (z - z_i)}$$

is initially defined on $\Omega - \{z_1, \dots, z_n\}$, and is bounded near each z_i . Thus z_i is a removable singularity of $g(z)$. One can define $g(z)$ at z_i to be the limit of $g(z)$ at z_i . Thus you can write,

$$f(z) = \prod_{i=1}^N (z - z_i) g(z)$$

where $g(z)$ is holomorphic on Ω and nowhere vanishing in the closure of D_R . Thus it suffices to prove Jensen's formula for functions like $g(z)$, ie holomorphic on Ω and nowhere vanishing on closure of D_R , and functions of the form $(z - z_j)$.

Step 3: Since g is nonvanishing on D_r which simply connected and open, by Theorem 6.2 in Chapter 3 of Stein, $\exists h = \log(g)$ or equivalently $g(x) = e^{h(x)}$.

Let $h(z) = u(z) + iv(z)$

$$\begin{aligned} |g(z)| &= \left| e^{h(z)} \right| \\ &= \left| e^{u(z) + iv(z)} \right| \\ &= \left| e^{u(z)} e^{iv(z)} \right| \\ &= e^{u(z)} \end{aligned}$$

Thus $\log |g(z)| = u(z)$. By the mean value property (Corollary 7.3 in Chapter 2)

$$\begin{aligned}\log |g(0)| = u(0) &= \frac{1}{2\pi} \int_0^{2\pi} u(Re^{i\theta}) d\theta \\ &= \frac{1}{2\pi} \int_0^{2\pi} \log |g(Re^{i\theta})| d\theta\end{aligned}$$

Since $g(z)$ is nonvanishing on D_R it has no zeros in D_R so the first summation is 0 and g satisfies the theorem.

Step 4: Now that we proved that for $g(z)$ nonvanishing on D_r , we now reduce to the case of $z - a$ for $a \in D_r$.

Since $|a|/R \in \mathbb{R} \implies \log\left(\frac{|a|}{R}\right) = \log |a| - \log R$. Similarly $\log |Re^{i\theta} - a| = \log |R(e^{i\theta} - \frac{a}{R})| = \log R + \log |e^{i\theta} - \frac{a}{R}|$. Let $b = \frac{a}{R}$. Note $|b| = \frac{|a|}{R} < 1$. Thus $\log |Re^{i\theta} - a| = \log R - \log |e^{i\theta} - b|$
Thus,

$$\begin{aligned}\log\left(\frac{|a|}{R}\right) + \frac{1}{2\pi} \int_0^{2\pi} \log |Re^{i\theta} - a| d\theta &= \log |a| - \log R + \frac{1}{2\pi} \int_0^{2\pi} [\log R - \log |e^{i\theta} - b|] d\theta \\ &= \log |a| + \frac{1}{2\pi} \int_0^{2\pi} \log |e^{i\theta} - b| d\theta \\ &= \log |a| + \frac{1}{2\pi} \int_0^{2\pi} \log |e^{i\theta}(1 - e^{-i\theta}b)| d\theta \\ &= \log |a| + \frac{1}{2\pi} \int_0^{2\pi} \log |1 - e^{i\theta}b| d\theta\end{aligned}$$

In Homework 3 where we solved Exercise 11b from Chapter 3 in Stein, we proved that when $|b| < 1$, $\int_0^{2\pi} \log |1 - e^{i\theta}b| d\theta = 0$

Thus,

$$\log\left(\frac{|a|}{R}\right) + \frac{1}{2\pi} \int_0^{2\pi} \log |Re^{i\theta} - a| d\theta = \log |a| = \log(|0 - a|)$$

Thus theorem applies for $w - a$. By step 2, since the theorem applies for g and $w - a$ it applies for all functions which are the product which end up being all valid f . $\square$

$\Theta(z | \tau) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau} e^{2\pi i n z}$ and $\Pi(z | \tau) = \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{2\pi i z})(1 + q^{2n-1} e^{-2\pi i z})$ where $q = e^{2\pi i \tau}$

Claim 2. For all $z \in \mathbb{C}$ and $\tau \in \mathbb{H}$ we have the identity $\Theta(z | \tau) = \Pi(z | \tau)$

Proof. Fix $\tau \in \mathbb{H}$ and let $\Theta(z) = \Theta(z | \tau)$ and let $\Pi(z) = \Pi(z | \tau)$.

Look at the function $F(z) = \Theta(z)/\Pi(z)$. By proposition 1.1 and 1.2 in Chapter 10, $\Theta(z)$ and $\Pi(z)$ are entire with roots at $\frac{1}{2} + \frac{\tau}{2} + n + m\tau$ for $n, m \in \mathbb{Z}$ and all roots are simple zeros. Thus $F(z)$ is

entire. By Proposition 1.1 and 1.2 in Chapter 10, $\Theta(z)$ and $\Pi(z)$ have periods of 1 and τ . Thus F is constant, by Liouville's Theorem. Let $c = F(z)$. Note c is dependent on τ .

We want to show that $c = 1$.

Lets put $z = \frac{1}{2}$ into $\Theta(z) = c\Pi(z)$.

$$\begin{aligned} \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau} e^{\pi i n} &= c \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{2n-1} e^{\pi i})(1 + q^{2n-1} e^{-\pi i}) \\ &= \sum_{n \in \mathbb{Z}} (-1)^n q^{n^2} = c \prod_{n=1}^{\infty} (1 - q^{2n})(1 - q^{2n-1})(1 - q^{2n-1}) \\ &= c \prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1}) \text{ (Multiply neighboring values)} \end{aligned}$$

thus,

$$c = \frac{\sum_{n \in \mathbb{Z}} (-1)^n q^{n^2}}{\prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1})}$$

Evaluate at $z = \frac{1}{4}$,

$$\Theta\left(\frac{1}{4}\right) = \sum_{n \in \mathbb{Z}} e^{\pi i n^2 \tau} e^{\pi i \frac{n}{2}} = \sum_{n \in \mathbb{Z}} q^{n^2} i^n$$

Since $\frac{1}{i} = -i$, all odd terms are canceled,

$$\Theta\left(\frac{1}{4}\right) = \sum_{n \in \mathbb{Z}} q^{4n^2} (-1)^n$$

$$\begin{aligned} \Pi\left(\frac{1}{4}\right) &= \prod_{n=1}^{\infty} (1 - q^{2n})(1 + i q^{2n-1})(1 - i q^{2n-1}) \\ &= c \prod_{n=1}^{\infty} (1 - q^{2n})(1 + q^{4n-2}) \\ &= c \prod_{m=1}^{\infty} (1 - q^{4m})(1 - q^{8n-4}) \end{aligned}$$

There are 2 cases here $2m = 4n - 4$ and $2m = 4n - 2$.

thus

$$c = \frac{\sum_{n \in \mathbb{Z}} (-1)^n q^{n^2}}{\prod_{n=1}^{\infty} (1 - q^n)(1 - q^{2n-1})} = \frac{\sum_{n \in \mathbb{Z}} q^{4n^2} (-1)^n}{\prod_{m=1}^{\infty} (1 - q^{4m})(1 - q^{8n-4})}$$

The second half of the equality we could have also gotten by taking 4τ . Thus $c_\tau = c_{4\tau} = c(4^k t)$ for all $k \in \mathbb{N}$. Since $q^{4^k} = e^{i\pi 4^k \tau} \rightarrow 0$ as $k \rightarrow \infty$, by 2 we know $c(\tau) = 1$. Thus $\Theta(z) = \Pi(z)$. Thus $\Theta(z | \tau) = \Pi(z | \tau)$.

□